

Data Source & Scope Justification

The source data for this project is provided from the **ContosoRetailDW** database. It is important to note that this source is not a raw operational (OLTP) system, but rather an existing dimensional data warehouse containing pre-structured Fact and Dimension tables. For the purpose of this project, these tables were treated as the operational source system, and a complete independent ETL pipeline (Staging → Validation → Transformation → Data Warehouse) was designed and built around them, rather than reusing the source's existing structure as-is.

Out of the full set of tables available in ContosoRetailDW, four Fact tables were selected to represent the core business processes required by the project brief: **FactSales** (in-store sales transactions, also covering the Orders/Purchases requirement since the source does not maintain a separate Orders process), **FactOnlineSales** (online sales transactions, linked to individual customers), **FactInventory** (daily inventory snapshots, representing Stock Movements), and **FactExchangeRate** (daily currency exchange rates, representing financial Transactions). Each of these Fact tables was chosen because it maps directly to one of the business entities defined in the project scope (Sales, Orders/Purchases, Transactions, Stock Movements), while also offering a distinct data grain and a different set of data quality challenges — which allows the validation and transformation logic to be demonstrated in a meaningfully varied way.

Two Fact tables, **FactITSLA** and **FactITMachine**, were deliberately excluded from the scope, as they represent IT infrastructure operations (machine downtime and maintenance costs) rather than retail business processes, and therefore fall outside the defined project scope.

To support these four Fact tables, ten related Dimension tables were included based on their direct foreign key relationships: **DimDate**, **DimStore**, **DimProduct** (with its parent tables **DimProductSubcategory** and **DimProductCategory**), **DimPromotion**, **DimCurrency**, **DimChannel**, **DimCustomer**, and **DimGeography** (required by both DimStore and DimCustomer). Other available dimensions — **DimAccount**, **DimScenario**, **DimEmployee**, **DimEntity**, **DimSalesTerritory**, **DimMachine**, and **DimOutage** — were excluded from the core scope, as they are either unrelated to the selected Fact tables through any foreign key relationship, or belong to the IT/financial-reporting subject area that was already excluded above.